

The Closed-Disk Model of the Elliptic Plane

Projections and Constructed Objects

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August 27, 2026

Abstract

This note collects the mathematics behind the interactive viewer in this repository. It fixes the model — the sphere with antipodal points identified, drawn as a closed disk — gives the two projections the viewer offers, and then derives, for every object the program constructs, both its description on the sphere and the curve it becomes on the page. The point of the derivations is practical: knowing that the image of a line is a conic with a computable centre and computable semi-axes is what lets the drawing and the GCLC export emit one exact arc instead of a few hundred short chords.

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1 The model

1.1 The elliptic plane

Definition 1.1. The *elliptic plane* is the quotient

$$\mathbb{RP}^2 = \mathbb{S}^2 / \{\pm 1\}, \quad \mathbb{S}^2 = \{\mathbf{v} \in \mathbb{R}^3 : \|\mathbf{v}\| = 1\},$$

in which a point is an antipodal pair $[\mathbf{v}] = \{\mathbf{v}, -\mathbf{v}\}$, and a *line* is the image of a great circle. The metric is inherited from the round sphere.

Two facts govern everything that follows. First, two distinct points lie on exactly one line and two distinct lines meet in exactly one point: *there are no parallels*, and point and line play symmetric roles. Second, because the identification halves the sphere, every distance is at most $\pi/2$ and the total area is 2π rather than 4π .

1.2 Representatives and the closed disk

Each class $[\mathbf{v}]$ has a representative on the closed upper hemisphere

$$\mathbb{S}_+^2 = \{\mathbf{v} \in \mathbb{S}^2 : v_3 \geq 0\},$$

unique except on the equator $v_3 = 0$, where $\mathbf{v}$ and $-\mathbf{v}$ both lie in $\mathbb{S}_+^2$ and are the same elliptic point. Write

$$\text{up}(\mathbf{v}) = \begin{cases} -\mathbf{v}, & v_3 < 0, \\ \mathbf{v}, & v_3 \geq 0, \end{cases} \quad (1)$$

for the choice of that representative.

Remark 1.2 (why the code compares against $-\varepsilon$ and not 0). In the program (1) flips only when $v_3 < -\varepsilon$. On the equator the two representatives are equally correct, and a sampled curve whose endpoint lands at $v_3 = -10^{-17}$ must not be thrown to the other side of the disk; the tolerance keeps a curve that ends on the rim in one piece.

A *projection* is a homeomorphism $\pi: \mathbb{S}_+^2 \rightarrow \overline{\mathbb{D}}$ onto the closed unit disk $\overline{\mathbb{D}} = \{\mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\| \leq 1\}$. The picture is then the disk with *diametrically opposite boundary points identified*,

$$\mathbf{u} \sim -\mathbf{u} \quad (\|\mathbf{u}\| = 1),$$

which is the residue of the antipodal identification and the reason the rim is drawn dashed. Both projections below fix the rim pointwise, so this identification — and therefore the model, the picking and every measurement — is the same in either view.

2 The projections

2.1 Orthogonal projection

Definition 2.1. *Orthogonal* (vertical, “seen straight down”) projection is

$$\pi_o(\mathbf{v}) = (v_1, v_2), \quad \pi_o^{-1}(x, y) = (x, y, \sqrt{1 - x^2 - y^2}). \quad (2)$$

It is a bijection $\mathbb{S}_+^2 \rightarrow \overline{\mathbb{D}}$ and fixes the rim. It is neither conformal nor area-preserving; it compresses radially near the boundary, which is why a right angle drawn far from the centre visibly leans.

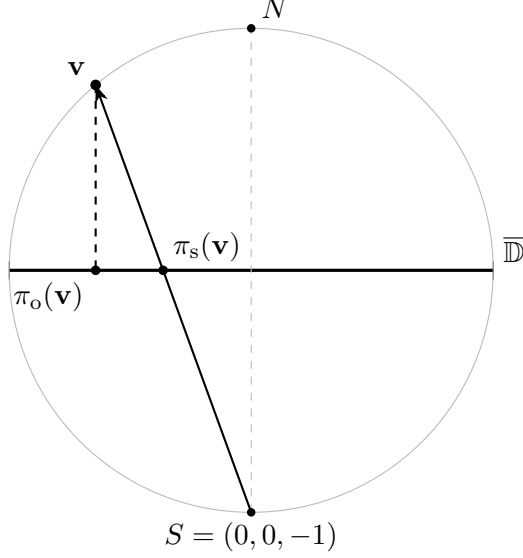


Figure 1: The two projections in a vertical cross-section. Orthogonal projection drops the point straight down; stereographic projection follows the ray from the south pole. Both send the closed upper hemisphere onto the closed unit disk and both fix the rim.

2.2 Stereographic projection

Definition 2.2. *Stereographic projection from the south pole $S = (0, 0, -1)$ is*

$$\pi_s(\mathbf{v}) = \frac{(v_1, v_2)}{1 + v_3}, \quad \pi_s^{-1}(x, y) = \frac{(2x, 2y, 1 - x^2 - y^2)}{1 + x^2 + y^2}. \quad (3)$$

Lemma 2.3. π_s maps $\mathbb{S}_+^2$ bijectively onto $\overline{\mathbb{D}}$ and fixes the rim pointwise.

Proof. For $\mathbf{v} \in \mathbb{S}^2$ with $\|\pi_s(\mathbf{v})\|^2 = (1 - v_3^2)/(1 + v_3)^2 = (1 - v_3)/(1 + v_3)$, which is a decreasing function of v_3 equal to 1 at $v_3 = 0$ and to 0 at $v_3 = 1$. Hence $v_3 \geq 0 \iff \|\pi_s(\mathbf{v})\| \leq 1$, and $v_3 = 0$ gives $\pi_s(\mathbf{v}) = (v_1, v_2)$, the identity on the equator. Formula (3) for π_s^{-1} is verified by substitution. $\square$

Remark 2.4. The inverse in (3) is a *rational* map, whereas π_o^{-1} involves $\sqrt{1 - x^2 - y^2}$, whose derivative blows up at the rim. This is a real numerical difference; see §10.

2.3 The metric each one induces

Pulling the round metric of $\mathbb{S}^2$ back through π^{-1} turns the disk into a metric object; the two answers are quite different and explain what each view is good for.

Proposition 2.5 (orthogonal). *With $r^2 = x^2 + y^2$ and (x, y) polar as $x = r \cos \theta$, $y = r \sin \theta$,*

$$ds_o^2 = dx^2 + dy^2 + \frac{(x dx + y dy)^2}{1 - x^2 - y^2} = \frac{dr^2}{1 - r^2} + r^2 d\theta^2, \quad dA_o = \frac{dx dy}{\sqrt{1 - x^2 - y^2}}. \quad (4)$$

Substituting $r = \sin \rho$ gives $ds_o^2 = d\rho^2 + \sin^2 \rho d\theta^2$: the radial coordinate r is the sine of the true distance ρ from the centre, and the rim $r = 1$ is at distance exactly $\pi/2$.

Proof. Differentiate $z = \sqrt{1 - x^2 - y^2}$ to get $dz = -(x dx + y dy)/z$ and substitute into $ds^2 = dx^2 + dy^2 + dz^2$. In polar coordinates $x dx + y dy = r dr$, so the extra term is $r^2 dr^2/(1 - r^2)$ and $dr^2 + r^2 dr^2/(1 - r^2) = dr^2/(1 - r^2)$. $\square$

Proposition 2.6 (stereographic).

$$ds_s^2 = \frac{4(dx^2 + dy^2)}{(1 + x^2 + y^2)^2}, \quad dA_s = \frac{4 dx dy}{(1 + x^2 + y^2)^2}. \quad (5)$$

The metric is a positive multiple of the Euclidean one, so π_s is conformal: an angle measured on the page is the angle of the geometry, everywhere and exactly.

That single line is the whole trade-off. In the orthogonal view distances read off the page by a fixed rule ($\rho = \arcsin r$ from the centre) but angles are distorted; in the conformal view angles are honest but distance is not read off the page at all.

3 Points

Definition 3.1. A *point* of the construction is an elliptic point $[\mathbf{v}]$, stored by its representative $\mathbf{v} \in \mathbb{S}_+^2$, $\|\mathbf{v}\| = 1$.

A click at (x, y) becomes $\mathbf{v} = \pi^{-1}(x, y)$ for whichever projection π is in force, after clamping outside points onto the rim:

$$\text{clamp}(x, y) = \begin{cases} (x, y)/r, & r > 1, \\ (x, y), & r \leq 1, \end{cases} \quad r = \sqrt{x^2 + y^2}. \quad (6)$$

Points are of two kinds. A *free* point carries its own position and can be dragged. A *derived* point carries a rule — a meet (§6.1) or a pole (§6) — and recomputes its vector from its parents each time it is asked, which is what makes the whole figure move when one vertex is dragged.

4 Lines

Definition 4.1. The *line* through two distinct points $[\mathbf{p}], [\mathbf{q}]$ is the great circle

$$\ell_{\mathbf{n}} = \{\mathbf{v} \in \mathbb{S}^2 : \mathbf{n} \cdot \mathbf{v} = 0\}, \quad \mathbf{n} = \frac{\mathbf{p} \times \mathbf{q}}{\|\mathbf{p} \times \mathbf{q}\|},$$

and $\mathbf{n}$ is defined up to sign, i.e. it is itself an elliptic point. The cross product vanishes exactly when $[\mathbf{p}] = [\mathbf{q}]$, in which case infinitely many lines pass through them and the construction refuses.

Writing $\boldsymbol{\nu} = (n_1, n_2)$ and $\hat{\boldsymbol{\nu}} = \boldsymbol{\nu}/\|\boldsymbol{\nu}\|$, so that $\|\boldsymbol{\nu}\|^2 = 1 - n_3^2$, we can say exactly what a line looks like on the page.

4.1 Orthogonal image: half an ellipse

Proposition 4.2. Let $|n_3| < 1$ and $\|\boldsymbol{\nu}\| > 0$. Then $\pi_o(\ell_{\mathbf{n}})$ is half of the ellipse centred at the origin with

$$\text{semi-major axis } 1 \text{ along } \hat{\boldsymbol{\nu}}^\perp = \frac{(-n_2, n_1)}{\|\boldsymbol{\nu}\|}, \quad \text{semi-minor axis } |n_3| \text{ along } \hat{\boldsymbol{\nu}}.$$

The half drawn is the one containing $-\text{sgn}(n_3)|n_3|\hat{\boldsymbol{\nu}}$, and it joins the antipodal rim points $\pm\hat{\boldsymbol{\nu}}^\perp$.

Proof. A point of the disk is on the image iff $n_1x + n_2y + n_3\sqrt{1 - x^2 - y^2} = 0$. Squaring,

$$n_3^2(1 - x^2 - y^2) = (n_1x + n_2y)^2 \iff \mathbf{u}^\top(\boldsymbol{\nu}\boldsymbol{\nu}^\top + n_3^2 I)\mathbf{u} = n_3^2, \quad \mathbf{u} = (x, y)^\top.$$

The matrix $M = \boldsymbol{\nu}\boldsymbol{\nu}^\top + n_3^2 I$ has eigenvector $\hat{\boldsymbol{\nu}}$ with eigenvalue $\|\boldsymbol{\nu}\|^2 + n_3^2 = 1$ and eigenvector $\hat{\boldsymbol{\nu}}^\perp$ with eigenvalue n_3^2 . In those coordinates, $\mathbf{u} = s\hat{\boldsymbol{\nu}} + t\hat{\boldsymbol{\nu}}^\perp$, the equation reads $s^2 + n_3^2 t^2 = n_3^2$, i.e.

$$\frac{s^2}{n_3^2} + \frac{t^2}{1} = 1.$$

So the semi-axis along $\hat{\boldsymbol{\nu}}$ is $|n_3| \leq 1$ and the one along $\hat{\boldsymbol{\nu}}^\perp$ is 1. Squaring doubled the solution set: only the half with $n_1 x + n_2 y$ and n_3 of opposite sign survives, and that half is bounded by $t = \pm 1$, $s = 0$, the two rim points $\pm \hat{\boldsymbol{\nu}}^\perp$. Finally, the point of $\ell_{\mathbf{n}}$ of greatest height is $(\hat{\mathbf{z}} - n_3 \mathbf{n}) / \|\hat{\mathbf{z}} - n_3 \mathbf{n}\|$, whose horizontal part is $-n_3 \boldsymbol{\nu} / \|\boldsymbol{\nu}\|$; this is the apex of the drawn half. $\square$

Corollary 4.3 (degenerate cases). $n_3 = 0$: the ellipse collapses onto its major axis and the line is drawn as the diameter along $\hat{\boldsymbol{\nu}}^\perp$. $|n_3| = 1$ (the equator): the line is the whole rim circle. These are exactly the two cases in which the program returns no conic and draws something else.

4.2 Stereographic image: a circular arc

Proposition 4.4. Let $n_3 \neq 0$. Then $\pi_s(\ell_{\mathbf{n}})$ is the arc, inside $\overline{\mathbb{D}}$, of the circle

$$\|\mathbf{u} - \mathbf{c}\| = \varrho, \quad \mathbf{c} = \frac{(n_1, n_2)}{n_3}, \quad \varrho = \frac{1}{|n_3|}. \quad (7)$$

If $n_3 = 0$ the image is the diameter along $\hat{\boldsymbol{\nu}}^\perp$, as before.

Proof. Substitute (3) into $\mathbf{n} \cdot \mathbf{v} = 0$. With $q = x^2 + y^2$,

$$\frac{2n_1 x + 2n_2 y + n_3(1 - q)}{1 + q} = 0 \iff n_3 q - 2n_1 x - 2n_2 y - n_3 = 0.$$

For $n_3 \neq 0$ divide by n_3 and complete the square: $\|\mathbf{u} - \mathbf{c}\|^2 = 1 + \|\mathbf{c}\|^2 = (n_3^2 + n_1^2 + n_2^2)/n_3^2 = 1/n_3^2$. For $n_3 = 0$ the equation is linear, $n_1 x + n_2 y = 0$. $\square$

Corollary 4.5 (the identification is visible). The circle (7) meets the unit circle where $\mathbf{c} \cdot \mathbf{u} = 0$, i.e. at the two antipodal rim points $\pm \hat{\mathbf{c}}^\perp = \pm \hat{\boldsymbol{\nu}}^\perp$ — the same pair as in Proposition 4.2.

Proof. Subtract $\|\mathbf{u}\|^2 = 1$ from $\|\mathbf{u} - \mathbf{c}\|^2 = 1 + \|\mathbf{c}\|^2$ to get $-2\mathbf{c} \cdot \mathbf{u} = 0$. $\square$

4.3 Parametrisation, and which half is drawn

For sampling and for cutting segments the program uses a parametrisation rather than the implicit equations. Choose an orthonormal basis $\mathbf{e}_1, \mathbf{e}_2$ of $\mathbf{n}^\perp$ and set

$$\gamma(t) = \cos t \mathbf{e}_1 + \sin t \mathbf{e}_2. \quad (8)$$

Its height is

$$\gamma_3(t) = e_{1,3} \cos t + e_{2,3} \sin t = R \cos(t - \varphi), \quad \varphi = \text{atan2}(e_{2,3}, e_{1,3}), \quad R = \sqrt{1 - n_3^2}, \quad (9)$$

so $\gamma_3 \geq 0$ precisely on the half-turn

$$t \in [\varphi - \frac{\pi}{2}, \varphi + \frac{\pi}{2}]. \quad (10)$$

That interval is the line: a half-turn, mapping onto half the ellipse (or onto the circular arc). The equator, $R = 0$, is the exception and takes the full turn $[0, 2\pi]$.

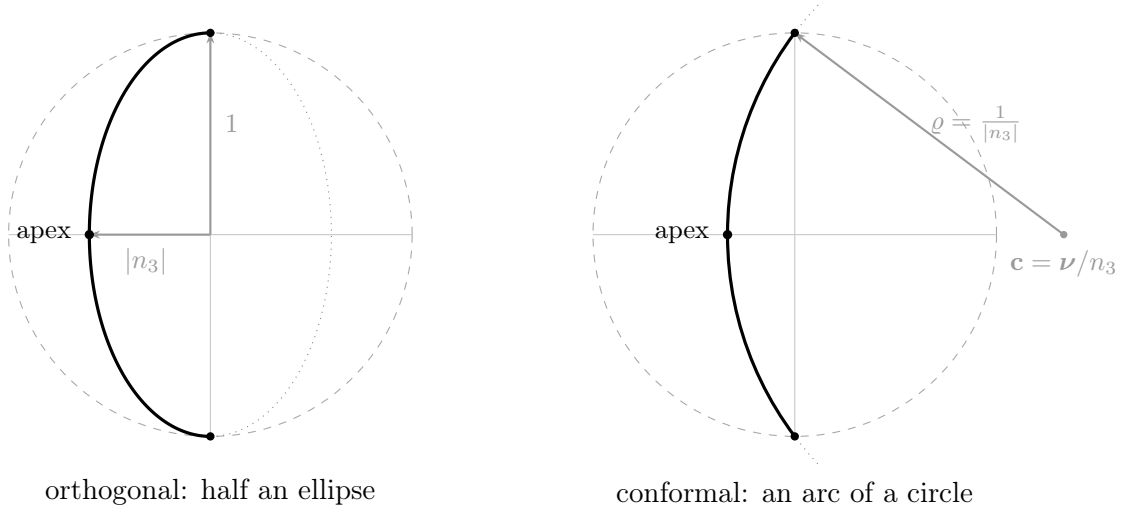


Figure 2: One and the same elliptic line, $\mathbf{n} = (0.8, 0, 0.6)$, drawn both ways. Both curves run between the same pair of antipodal rim points $\pm \hat{\nu}^\perp = (0, \pm 1)$ (Corollary 4.5), and the two marked apexes are the images of the same point $(-0.6, 0, 0.8)$ of the sphere. The dotted continuation on the left is the half of the ellipse that the lower hemisphere would project to; it is *not* part of the line.

5 Distance and segments

Definition 5.1. The *distance* between two elliptic points is the angle between their representatives, folded by the identification:

$$d([\mathbf{p}], [\mathbf{q}]) = \arccos(|\mathbf{p} \cdot \mathbf{q}|) \in [0, \frac{\pi}{2}]. \quad (11)$$

The absolute value is the whole content of the identification: whichever of $\pm \mathbf{q}$ is nearer to $\mathbf{p}$ is the one that counts, and no two points are ever more than a quarter turn apart. The rim is at distance exactly $\pi/2$ from the centre.

5.1 The shortest segment, and why it can arrive in two pieces

Let $\mathbf{p}, \mathbf{q}$ lie on $\ell_{\mathbf{n}}$ with parameters t_p, t_q in (8). Reduce

$$\delta \equiv (t_q - t_p + \pi) \bmod 2\pi - \pi \in (-\pi, \pi], \quad \delta \leftarrow \delta - \text{sgn}(\delta) \pi \quad \text{if } |\delta| > \frac{\pi}{2}, \quad (12)$$

the second step choosing the antipodal representative of $\mathbf{q}$ when it is the nearer one. The segment is then γ on $[t_p, t_p + \delta]$, of length $|\delta| = d(\mathbf{p}, \mathbf{q})$.

This arc need not stay in the upper hemisphere. By (9) the height vanishes at $t = \varphi \pm \pi/2$; since $|\delta| \leq \pi/2$ the arc crosses at most one of those, at a parameter t_* strictly between its ends. Splitting there and applying up to each piece — which on an arc means replacing $(\mathbf{e}_1, \mathbf{e}_2)$ by $(-\mathbf{e}_1, -\mathbf{e}_2)$ — gives *one or two* drawn arcs. Geometrically: the shortest route leaves the disk through the rim and re-enters at the diametrically opposite boundary point.

Remark 5.2. The cut is computed from (9) analytically, not detected from the samples, so the two pieces do not depend on how finely the curve is drawn — and the export can emit them as two exact arcs.

6 Duality: pole and polar

Definition 6.1. The *polar* of a point $[\mathbf{p}]$ is the line $\ell_{\mathbf{p}}$ with normal $\mathbf{p}$; the *pole* of a line $\ell_{\mathbf{n}}$ is the point $[\mathbf{n}]$.

Proposition 6.2. $\ell_{\mathbf{p}} = \{[\mathbf{v}] : d([\mathbf{p}], [\mathbf{v}]) = \pi/2\}$, and pole and polar are mutually inverse: the pole of the polar of $[\mathbf{p}]$ is $[\mathbf{p}]$.

Proof. $d([\mathbf{p}], [\mathbf{v}]) = \pi/2 \iff |\mathbf{p} \cdot \mathbf{v}| = 0 \iff \mathbf{v} \in \ell_{\mathbf{p}}$. The second claim is immediate from the definitions, both sides being the class $[\mathbf{p}]$. $\square$

So a point and a line are the same kind of object; the viewer offers them as one tool. Dualising a whole figure exchanges “lies on” with “passes through”, collinear points with concurrent lines.

6.1 Meet

Proposition 6.3. Two distinct lines $\ell_{\mathbf{n}_1}, \ell_{\mathbf{n}_2}$ meet in exactly one point,

$$[\mathbf{n}_1 \times \mathbf{n}_2]. \quad (13)$$

Proof. $\mathbf{v}$ lies on both iff it is orthogonal to $\mathbf{n}_1$ and $\mathbf{n}_2$, i.e. spans $(\text{span}\{\mathbf{n}_1, \mathbf{n}_2\})^\perp$, a line of $\mathbb{R}^3$ when $\mathbf{n}_1 \neq \pm \mathbf{n}_2$; its two unit vectors are one elliptic point. $\square$

This is exactly dual to $\mathbf{n} = \mathbf{p} \times \mathbf{q}$ for the join, and it is why the elliptic plane has no parallels.

6.2 Perpendiculars

Proposition 6.4. Let $[\mathbf{p}]$ be a point and $\ell_{\mathbf{n}}$ a line with $[\mathbf{p}] \neq [\mathbf{n}]$. There is exactly one line through $[\mathbf{p}]$ perpendicular to $\ell_{\mathbf{n}}$, namely

$$\ell_{\mathbf{m}}, \quad \mathbf{m} = \frac{\mathbf{p} \times \mathbf{n}}{\|\mathbf{p} \times \mathbf{n}\|}, \quad (14)$$

which is the join of $[\mathbf{p}]$ and the pole $[\mathbf{n}]$. Its foot — the point of $\ell_{\mathbf{n}}$ nearest $[\mathbf{p}]$ — is

$$[\mathbf{n} \times \mathbf{m}], \quad (15)$$

and

$$d([\mathbf{p}], \ell_{\mathbf{n}}) = \arcsin(|\mathbf{p} \cdot \mathbf{n}|) = \frac{\pi}{2} - d([\mathbf{p}], [\mathbf{n}]). \quad (16)$$

If $[\mathbf{p}] = [\mathbf{n}]$ — the point is the pole — then every line through $[\mathbf{p}]$ is perpendicular to $\ell_{\mathbf{n}}$ and (14) degenerates.

Proof. Two lines are perpendicular iff their normals are orthogonal, $\mathbf{n} \cdot \mathbf{m} = 0$; and $\ell_{\mathbf{m}}$ passes through $[\mathbf{p}]$ iff $\mathbf{p} \cdot \mathbf{m} = 0$. Both hold for $\mathbf{m} \propto \mathbf{p} \times \mathbf{n}$, and that vector is determined up to sign whenever $\mathbf{p} \times \mathbf{n} \neq 0$. Since $\mathbf{m} \perp \mathbf{n}$, the pole $[\mathbf{n}]$ lies on $\ell_{\mathbf{m}}$: every line through the pole of $\ell_{\mathbf{n}}$ crosses it at a right angle. Equation (15) is (13) applied to $\ell_{\mathbf{n}}$ and $\ell_{\mathbf{m}}$. For (16), decompose $\mathbf{p} = (\mathbf{p} \cdot \mathbf{n})\mathbf{n} + \mathbf{p}^\parallel$ with $\mathbf{p}^\parallel \in \mathbf{n}^\perp$; the nearest point of $\ell_{\mathbf{n}}$ is $[\mathbf{p}^\parallel]$ and $\cos d = \|\mathbf{p}^\parallel\| = \sqrt{1 - (\mathbf{p} \cdot \mathbf{n})^2}$. $\square$

Corollary 6.5. All perpendiculars to a given line are concurrent — they meet at its pole. Dropping perpendiculars from two different points onto one line and watching them cross at the pole is the cleanest check that the construction is right.

7 Angles, triangles and area

7.1 Tangents and the angle at a vertex

For $[\mathbf{a}] \neq [\mathbf{b}]$ let $\mathbf{b}^* = \text{sgn}(\mathbf{a} \cdot \mathbf{b})\mathbf{b}$ be the nearer representative. The unit tangent at $\mathbf{a}$ along the shortest arc towards $[\mathbf{b}]$ is

$$\mathbf{t}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{b}^* - (\mathbf{a} \cdot \mathbf{b}^*)\mathbf{a}}{\|\mathbf{b}^* - (\mathbf{a} \cdot \mathbf{b}^*)\mathbf{a}\|}, \quad (17)$$

and the angle at $[\mathbf{a}]$ in the “triangle” $\mathbf{abc}$ is

$$\alpha = \arccos(\mathbf{t}(\mathbf{a}, \mathbf{b}) \cdot \mathbf{t}(\mathbf{a}, \mathbf{c})) \in [0, \pi]. \quad (18)$$

At exactly $d = \pi/2$ both representatives of $\mathbf{b}$ are equally near and the direction is defined only up to sign.

7.2 Girard’s theorem

Proposition 7.1 (Girard). *A triangle whose three shortest sides bound a disk has area*

$$\text{Area} = \alpha + \beta + \gamma - \pi. \quad (19)$$

The excess is therefore not an error term but the area itself. It is positive for every genuine triangle — the angle sum always beats π — and it goes to 0 as the triangle shrinks, which is Euclidean geometry appearing in the limit.

7.3 When three points bound nothing

Not every triple bounds a triangle, and (19) must not be applied blindly.

Proposition 7.2. *Walk the three shortest sides $[\mathbf{a}] \rightarrow [\mathbf{b}] \rightarrow [\mathbf{c}] \rightarrow [\mathbf{a}]$, at each step lifting to the nearer representative of the far endpoint. The walk returns to $+\mathbf{a}$ if and only if*

$$(\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a}) > 0, \quad (20)$$

and to $-\mathbf{a}$ otherwise.

Proof. Set $\mathbf{b}_1 = \sigma_1 \mathbf{b}$ with $\sigma_1 = \text{sgn}(\mathbf{a} \cdot \mathbf{b})$. Then $\mathbf{c}_1 = \text{sgn}(\mathbf{b}_1 \cdot \mathbf{c}) \mathbf{c} = \sigma_1 \text{sgn}(\mathbf{b} \cdot \mathbf{c}) \mathbf{c}$, and finally $\mathbf{a}_1 = \text{sgn}(\mathbf{c}_1 \cdot \mathbf{a}) \mathbf{a} = \text{sgn}(\mathbf{a} \cdot \mathbf{b}) \text{sgn}(\mathbf{b} \cdot \mathbf{c}) \text{sgn}(\mathbf{c} \cdot \mathbf{a}) \mathbf{a}$, because the leading σ_1 appears twice and squares away. $\square$

When the product is negative the closed curve made of the three sides runs out through the rim and back in on the other side; it is a non-contractible loop in $\mathbb{RP}^2$, it bounds no disk, and Girard’s formula has nothing to apply to. The program reports that rather than printing a number, and pulling one vertex back in restores the area.

7.4 The marks that are drawn

Both angle marks are constructed *on the sphere* and then projected, so they are honest about the distortion of the view rather than faking a square corner.

Angle arc. With $\mathbf{t}_1 = \mathbf{t}(\mathbf{a}, \mathbf{b})$, $\mathbf{t}_2 = \mathbf{t}(\mathbf{a}, \mathbf{c})$, θ the angle (18) and $\mathbf{t}_\perp = \frac{\mathbf{t}_2 - (\mathbf{t}_1 \cdot \mathbf{t}_2)\mathbf{t}_1}{\|\mathbf{t}_2 - (\mathbf{t}_1 \cdot \mathbf{t}_2)\mathbf{t}_1\|}$, the mark at radius ρ (the program uses $\rho = 0.13$) is

$$s \mapsto \cos \rho \mathbf{a} + \sin \rho (\cos s \mathbf{t}_1 + \sin s \mathbf{t}_\perp), \quad s \in [0, \theta]. \quad (21)$$

Right-angle mark. At a foot $\mathbf{f}$ where $\ell_{\mathbf{n}_1}$ and $\ell_{\mathbf{n}_2}$ cross, the two unit tangents along the lines are $\mathbf{t}_i = \widehat{\mathbf{n}_i \times \mathbf{f}}$, and the corners

$$\widehat{\mathbf{f} + h\mathbf{t}_1}, \quad \mathbf{f} + \widehat{h(\mathbf{t}_1 + \mathbf{t}_2)}, \quad \widehat{\mathbf{f} + h\mathbf{t}_2} \quad (22)$$

(with $\widehat{}$ meaning normalise, then fold up by (1)) form a small square on the sphere. Under π_o it comes out as a slanted parallelogram except near the centre; under π_s it squares up by itself, because π_s is conformal. Both marks are dropped altogether when they would straddle the rim, where they would tear across the disk.

8 Circles

Definition 8.1. The *circle* of centre $[\mathbf{a}]$ and radius $r \in (0, \pi/2]$ is $\{[\mathbf{v}] : d([\mathbf{a}], [\mathbf{v}]) = r\}$, i.e. the plane section $\mathbf{a} \cdot \mathbf{v} = \cos r$ of the sphere, of Euclidean centre $\cos r \mathbf{a}$ and Euclidean radius $\sin r$.

At $r = \pi/2$ it degenerates to the polar line of its centre — in this geometry a circle of large enough radius *is* a line, which is the cleanest way to see the duality at work.

Proposition 8.2 (orthogonal image). *Let $\mathbf{e}_1, \mathbf{e}_2$ be an orthonormal basis of $\mathbf{a}^\perp$ and let $A = \sin r [\mathbf{e}_1^{xy} \ \mathbf{e}_2^{xy}] \in \mathbb{R}^{2 \times 2}$ be the matrix of their horizontal parts. Then π_o of the circle is the ellipse*

$$\mathbf{u}(t) = \cos r \mathbf{a}^{xy} + A \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, \quad (23)$$

centred at $\cos r \mathbf{a}^{xy}$, whose semi-axes are the singular values of A taken along its left singular vectors. If $A = U \Sigma V^\top$, the semi-axis vectors are $\sigma_1 \mathbf{u}_1$ and $\sigma_2 \mathbf{u}_2$.

Proof. Parametrise the plane section as $\mathbf{v}(t) = \cos r \mathbf{a} + \sin r (\cos t \mathbf{e}_1 + \sin t \mathbf{e}_2)$ and drop the third coordinate. The image of the unit circle under a linear map A is an ellipse whose principal axes are given by the singular value decomposition. $\square$

Proposition 8.3 (stereographic image). *If $a_3 + \cos r \neq 0$, then π_s of the circle is the circle*

$$\|\mathbf{u} - \mathbf{c}\| = \varrho, \quad \mathbf{c} = \frac{(a_1, a_2)}{a_3 + \cos r}, \quad \varrho = \frac{\sin r}{|a_3 + \cos r|}. \quad (24)$$

If $a_3 + \cos r = 0$ the circle passes through the south pole and its image is a straight line.

Proof. Substitute (3) into $\mathbf{a} \cdot \mathbf{v} = \cos r$. With $q = x^2 + y^2$ and $c = \cos r$,

$$2a_1x + 2a_2y + a_3(1 - q) = c(1 + q) \iff (a_3 + c)q - 2a_1x - 2a_2y - (a_3 - c) = 0.$$

Dividing by $a_3 + c$ and completing the square gives centre $(a_1, a_2)/(a_3 + c)$ and

$$\varrho^2 = \frac{a_1^2 + a_2^2}{(a_3 + c)^2} + \frac{a_3 - c}{a_3 + c} = \frac{a_1^2 + a_2^2 + a_3^2 - c^2}{(a_3 + c)^2} = \frac{1 - c^2}{(a_3 + c)^2} = \frac{\sin^2 r}{(a_3 + c)^2}. \quad \square$$

Setting $r = \pi/2$ in (24) gives $\mathbf{c} = (a_1, a_2)/a_3$ and $\varrho = 1/|a_3|$, which is exactly (7) with $\mathbf{n} = \mathbf{a}$: the circle of radius $\pi/2$ about $[\mathbf{a}]$ is the polar of $[\mathbf{a}]$, as it must be. That both propositions are special cases of one formula is why a single “draw a conic” routine serves lines and circles alike.

Like a segment, a circle need not stay in the upper hemisphere: when the distance of the centre from the middle of the disk plus the radius exceeds $\pi/2$, the part below the equator folds up by (1) and re-enters the disk on the far side. The height along the circle is $z(t) = \cos r a_3 + \sin r \sqrt{1 - a_3^2} \cos(t - \varphi)$, so the cut is at $t = \varphi \pm \arccos(-\cot r a_3 / \sqrt{1 - a_3^2})$, computed exactly; the folded piece is the same arc with $\mathbf{a}, \mathbf{e}_1, \mathbf{e}_2$ all negated, and its image lies on the conic of these propositions with $\mathbf{a}$ replaced by $-\mathbf{a}$.

9 The image of every object, side by side

Table 1 is the whole model in one place. In it $\boldsymbol{\nu} = (n_1, n_2)$, $\hat{\boldsymbol{\nu}}^\perp = (-n_2, n_1)/\|\boldsymbol{\nu}\|$, and every “conic” is given as (centre; semi-axis vectors), which is precisely the data a drawing command needs.

object	on the sphere	on the page
point	$\mathbf{v} \in \mathbb{S}_+^2, \ \mathbf{v}\ = 1$	$\pi_o(\mathbf{v}) = (v_1, v_2)$ $\pi_s(\mathbf{v}) = (v_1, v_2)/(1 + v_3)$
line	$\{\mathbf{v} : \mathbf{n} \cdot \mathbf{v} = 0\}, \mathbf{n} = \widehat{\mathbf{p} \times \mathbf{q}}$	o : half-ellipse $(\mathbf{0}; \hat{\nu}^\perp, n_3 \hat{\nu})$ s : circle $(\nu/n_3; 1/ n_3)$
diameter	$n_3 = 0$	straight, along $\hat{\nu}^\perp$ (both views)
equator	$\mathbf{n} = \pm \hat{\mathbf{z}}$	the rim circle (both views)
rim ends	$\pm \hat{\nu}^\perp$	antipodal boundary points, identified
segment	$\gamma _{[t_p, t_p + \delta]}, \delta \leq \pi/2$	one or two arcs of the above conic
distance	—	$d = \arccos \mathbf{p} \cdot \mathbf{q} \leq \pi/2$
pole	$[\mathbf{n}]$	a point; where all perpendiculars to $\ell_{\mathbf{n}}$ meet
polar	$\{\mathbf{v} : \mathbf{p} \cdot \mathbf{v} = 0\}$	the line of everything $\pi/2$ from $[\mathbf{p}]$
meet	$[\mathbf{n}_1 \times \mathbf{n}_2]$	always exists, always unique
perpendicular	$\ell_{\mathbf{m}}, \mathbf{m} = \widehat{\mathbf{p} \times \mathbf{n}}$	the join of $[\mathbf{p}]$ and $[\mathbf{n}]$
foot	$[\mathbf{n} \times \mathbf{m}]$	nearest point of $\ell_{\mathbf{n}}$ to $[\mathbf{p}]$
point–line distance	—	$\arcsin \mathbf{p} \cdot \mathbf{n} = \frac{\pi}{2} - d([\mathbf{p}], [\mathbf{n}])$
angle	(17)–(18)	distorted under o , true under s
triangle area	Girard	$\alpha + \beta + \gamma - \pi$, if (20) holds
circle $d = r$	$\mathbf{a} \cdot \mathbf{v} = \cos r$	o : ellipse, Prop. 8.2 s : circle (24)

Table 1: Every constructed object, on the sphere and in the two views. o = orthogonal, s = stereographic (conformal).

10 Numerical remarks

The rim is ill-conditioned under π_o . Lifting by (2) evaluates $\sqrt{1 - r^2}$, and near $r = 1$

$$\frac{\partial}{\partial r} \sqrt{1 - r^2} = \frac{-r}{\sqrt{1 - r^2}} \rightarrow -\infty.$$

A relative perturbation ε in r produces an error of order $\sqrt{\varepsilon}$ in v_3 : at double precision, roughly 10^{-8} rather than 10^{-16} . Round-trip identities that hold to 10^{-12} in the middle of the disk hold only to about 10^{-6} at the boundary, and tolerances must say so. The stereographic lift (3) is rational and has no such problem, which is a second, quieter reason to prefer the conformal view when working near the rim.

Signs on the equator. $\mathbf{n}$ and $-\mathbf{n}$ describe the same line and $\mathbf{v}, -\mathbf{v}$ the same point, so no formula may depend on the sign chosen. Where a sign *is* chosen — the drawn half of an ellipse, the nearer representative in (17), the fold in (1) — it is chosen from the geometry (greatest height, largest inner product), never from the storage.

The far end of the ellipse is not on the line. In Proposition 4.2 the full ellipse is the projection of the whole great circle, upper *and* lower half. The point $+\operatorname{sgn}(n_3)|n_3|\hat{\nu}$ — the apex of the other half — is therefore the image of a point of the lower hemisphere, and lifting it back with π_o^{-1} lands somewhere else entirely. Only half the conic is the line (Figure 2).

11 Correspondence with the implementation

Every formula above is realised by a named function, and the two agree numerically: the propositions of §2–§8 were checked against the code on randomly generated configurations.

formula	name in <code>elliptic/geometry.py</code>
(1)	<code>upper</code>
(2)	<code>Orthogonal.project</code> , <code>Orthogonal.lift</code>
(3)	<code>Stereographic.project</code> , <code>Stereographic.lift</code>
Prop. 4.2	<code>line_ellipse</code> , <code>Orthogonal.conic</code>
Prop. 4.4	<code>Stereographic.conic</code>
(8)–(10)	<code>line_arc</code> , <code>arc_vectors</code>
(11)	<code>distance</code>
segment split at $\gamma_3 = 0$	<code>segment_arcs</code>
(13)	<code>meet_vector</code>
polar / pole	<code>polar_normal</code> , <code>pole_vector</code>
(14)	<code>perpendicular_normal</code>
(15)	<code>foot_of_perpendicular</code>
(16)	<code>distance_to_line</code>
(17), (18)	<code>tangent</code> , <code>arc_angle</code> , <code>triangle_angles</code>
(20)	<code>bounds_a_disk</code>
(19)	<code>triangle_area</code>
Prop. 8.2 (the SVD)	<code>small_circle_ellipse</code>
(24)	<code>Stereographic.point_conic</code>
circle, cut at $z = 0$ and folded	<code>circle_arcs</code> , <code>circle_vector_paths</code>
angle / right-angle marks	<code>angle_arc</code> , <code>right_angle_marker</code>

The export to GCLC (Predrag Janičić, University of Belgrade) rests on exactly the propositions above: because the image of a line under either projection is a conic with a computable centre and computable semi-axes, each line is written as a single `drawellipsearc` and each segment as one or two `drawellipsearc2` commands, rather than as a sampled chain of short segments. The two degenerate cases — $n_3 = 0$ and the equator — are written as a straight segment and as the rim circle respectively.